

KSoR

Knowledge System of Record — Complete Guide

(Roman Urdu + English — panaversity/ksor repo ka poora tafseeli khulasa)

Ye notes GitHub repo **panaversity/ksor** ke README par mabni hain — poori tafseel ke saath, koi concept chhoote bina. KSoR ek open-source SDK hai jo **governed, authoritative knowledge systems** banata hai — insaanon aur AI agents dono ke liye.

■ Zaroori baat: ye package abhi **0.0.0** version par hai — sirf naam reserve karne ke liye. Neeche jo bhi commands/features bataye gaye hain, wo *design* hain, abhi *implement* nahi hue. Ye asal mein sirf naam bachane wala 'placeholder' hai.

Is Guide mein kya kya hai

- KSoR kyun banaya gaya — asal masla kya hai
- KSoR ki definition, aur Traditional SoR se farq
- KSoR sirf 'knowledge base' se zyada kyun hai
- 'One Source, Two Surfaces' ka usool
- 7 Core Principles
- Kya bana sakte hain KSoR se (Organizational, Domain, Product/Method)
- Ek misaal — Accounting KSoR
- Quick Start, CLI, aur Project Structure
- Knowledge as Code + Build Provenance
- Agent Surface (MCP) aur KSoR vs RAG vs CMS
- Governance Model aur Knowledge Boundaries
- AI-Native Architecture mein KSoR ka role
- Design Goals aur Project Status

1. KSoR Kyun Banaya Gaya

Enterprises dashak se **Systems of Record** use karte aa rahe hain — jaise accounting system (financial transactions ke liye authoritative), CRM (customer records ke liye), HRIS (employee records ke liye). Agar spreadsheet aur accounting ledger mein farq ho, to **ledger jeetta hai** — wahi authoritative hai. Ye systems is sawal ka jawab dete hain: **'Business ki asal operational state kya hai?'**

AI Agents ek naya masla laate hain — unhe ye bhi jaanna hota hai:

- Kaunsi policies lagu hoti hain?
- Is decision par kaunse rules govern karte hain?
- Kaunsi procedure follow karni chahiye?
- Organization is term ka kya matlab leta hai?
- Kaunse thresholds approved hain?
- Kaunsi methodology use honi chahiye?
- Kya exceptions maujood hain?
- Is jawab ke peeche kaunse sources hain?
- Agar jawab maloom na ho to agent kya kare?

Ye jaankari aksar bikhri hoti hai — documents, wikis, PDFs, slides, websites, manuals, repos, employee ke zehan mein, prompts mein, RAG indexes mein. Koi authoritative jawab nahi hota ke **'AI ko kaunsi knowledge par bharosa karna chahiye?'** — KSoR isi masle ko hal karne ke liye bana hai.

2. KSoR Ki Definition

Knowledge System of Record (KSoR) — wo authoritative, governed source hai jahan se insaan aur AI agents dono *samajhte*, *faisla karte*, aur *amal* karte hain. Ismein ye sab aa sakta hai: domain knowledge, policies, procedures, rules, standards, methods, definitions, decision criteria, thresholds, specifications, controls, examples, exceptions, workflows, provenance (asal source), aur supporting material.

■ Maqsad sirf information ko searchable banana nahi hai — maqsad ye establish karna hai: **'Ye wo knowledge hai jis se hum operate karte hain.'**

Traditional SoR vs Knowledge SoR

	Traditional System of Record	Knowledge System of Record
Asal maqsad	Operational state record karna	Institutional knowledge record karna
Content	Transactions, balances, customers, inventory	Rules, policies, methods, standards, definitions
Misaal systems	ERP, CRM, HRIS, accounting	KSoR
Bunyadi sawal	Abhi kya sach hai?	Hum kya jaante hain, kaise operate karna hai?
Kis ke liye behtar	Applications aur business processes	Insaan aur AI agents
Badalne ka tareeqa	Transactions	Review, governance, versioning
AI ka role	Sirf tool consumer	First-class knowledge consumer

Ek capable enterprise agent KSoR se policy padh sakta hai, CRM se current account data le sakta hai, governed rule apply kar sakta hai, action le sakta hai, aur result wapas traditional SoR mein likh sakta hai.

3. KSoR Sirf Knowledge Base Se Zyada Hai

Ek knowledge base sirf information **store** karta hai. KSoR **authority** establish karta hai — yehi farq zaroori hai.

Aam knowledge base sirf storage, search, retrieval, similarity, ya question-answering par focus karta hai. KSoR ko in sawalon ka bhi jawab dena hota hai:

- Is knowledge ka malik kaun hai?
- Ye kahan se aayi?
- Kaunsi version authoritative hai?
- Kya isay review kiya gaya?
- Iska scope kya hai?
- Jab sources aapas mein takrayen to kya hoga?
- Kya AI evidence aur apni guess mein farq kar sakta hai?
- Kya jawab wapas apne source tak trace ho sakta hai?
- Agar KSoR mein jawab hi na ho to kya hoga?

■ Isi liye KSoR **governance, provenance, citations, versioning, aur abstention (jawab na dena jab maloom na ho)** ko architecture ka zaroori hissa manta hai — optional feature nahi.

4. Ek Source, Do Surfaces

KSoR ka usool: **insaan aur AI agents ko organization ki knowledge ke alag-alag versions se kaam nahi karna chahiye**. Ek governed source do surfaces banata hai:

- **Human Surface** — ek padhne wali website: search/browse, read/learn, review/share
- **Agent Surface** — MCP ke zariye: search/retrieve, cite/reason, abstain/act

Website alag se maintain nahi hoti, aur agent corpus website ki chhupi hui copy nahi hai — dono ek hi **governed source** ke projections hain, isi liye 'Same Truth'.

5. Saat Core Principles

#	Principle	Matlab
1	One Authoritative Source	Knowledge ki ek hi canonical jagah honi chahiye — representations alag ho sakti hain, source ek hi rahe
2	Insaan aur Agents dono First-Class	Ab documentation sirf insaan ke liye nahi likhi jaati — AI assistants, agents, workflows sab consumers hain
3	Provenance Matters	Jawab kis document se aaya, kaunsi version, kab bana — ye chain hamesha maloom honi chahiye
4	Citation Before Confidence	AI ka confident lehja proof nahi hai — jawab ko trace hona chahiye, model ki yaadasht par nahi
5	Abstention Is a Feature	Agar KSoR mein jawab nahi hai, to sahi jawab hai: 'Ye maloom nahi' — guess karna nahi
6	Governance Before Retrieval	Behtareen search bhi ungoverned knowledge ko tez retrieve karega — pehla sawal ye ho ke kya knowledge mein jaana chahiye, kaun badal sakta hai
7	Vendor Neutrality	Aapki knowledge kisi AI company ki milkiyat nahi honi chahiye — ChatGPT, Claude, ya kisi bhi model se kaam aaye

6. KSoR Se Kya Bana Sakte Hain

KSoR jaan-boojh kar kisi khaas industry tak mehdood nahi — ye teen tarah ke KSoRs support karta hai:

Type	Misaalein
Organizational KSoR	Agent Factory KSoR, Engineering KSoR, Product Management KSoR, Security KSoR, AI Governance KSoR
Domain KSoR	Accounting, Government Contracting, Healthcare, Legal, Banking, Insurance, Supply Chain, Sales
Product/Method KSoR	Design System, API Standards, Architecture, Implementation Method, Compliance Framework

Vertical KSoR sirf KSoR ka ek application hai — kisi khaas profession/industry/domain ke liye authoritative knowledge layer.

Misaal — Accounting KSoR

Socho ek Accounting KSoR hai jismein revenue-recognition, capitalization, bad-debt policies; month-end close aur journal-entry-review procedures; segregation-of-duties aur approval-thresholds controls; glossary; aur examples hon.

Agar poocha jaye: *'Kya ye \$42,000 wala software implementation cost capitalize ho sakta hai?'* — agent organization ki capitalization policy nikalega, criteria apply karega, source cite karega, aur farq karega ke KSoR mein kya likha hai aur apna reasoning kya hai. Agar policy is situation ko cover nahi karti, to system khud se policy banayega nahi.

7. Quick Start aur CLI

■ Yaad rahe: neech diye commands abhi **design** hain — chalane par 'status notice' print hoga aur exit code 2 dega. Ye implement nahi hue.

```
npx @panaversity/ksor init my-ksor # Naya KSoR banana
cd my-ksor
npx @panaversity/ksor dev # Local development, website live update
npx @panaversity/ksor build # Deployable human surface banana
npx @panaversity/ksor serve # Agent surface (MCP) ke zariye expose karna
```

CLI vocabulary jaan-boojh kar chhota rakha gaya hai: `init`, `dev`, `build`, `serve` — bas ye chaar commands.

8. Project Structure

```
my-ksor/  
■■■ knowledge/ # authoritative corpus (plain Markdown)  
■ ■■■ about.md  
■ ■■■ policies/  
■ ■■■ procedures/  
■■■ site/ # human-surface config (Docusaurus)  
■■■ .agents/skills/ # coding agents ke liye instructions  
■■■ instance.md # is KSoR ki identity/maqsad
```

Folder/File	Kaam
knowledge/	Authoritative corpus — plain Markdown isliye ke portable, diffable, reviewable, version-controlled ho, insaan aur AI dono padh sakein, koi proprietary database na ho
instance.md	Is KSoR instance ki identity aur maqsad batata hai
site/	Human-readable surface ki config — ordinary source code rahega, koi opaque hosted service nahi
.agents/	AI coding agents ke liye instructions aur reusable skills — jaise 'policy add karo', 'provenance check karo', 'quiz banao'

9. Knowledge as Code + Build Provenance

KSoR institutional knowledge ko waise treat karta hai jaise software teams source code ko karti hain:

```
authored → reviewed → version controlled → validated → tested → built → published → consumed
```

Git sirf storage nahi rehta — history, authorship, diffs, branches, pull requests, approvals, releases, rollback, aur reproducible builds deta hai. Isi se ye shift mumkin hoti hai: **institutional knowledge governed infrastructure ban jati hai.**

Build Provenance — har jawab trace ho sake

Har production jawab wapas us knowledge tak traceable hona chahiye jisne usay banaya. Ek `build.lock.json` file included documents, unke hashes, source commit, KSoR version, waghera record karti hai. Isse ye chain banti hai:

```
AI Answer → Retrieved Passage → Knowledge Document → KSoR Build → Git Commit → Reviewed Source
```

Jab koi poochay 'agent ne ye kyun kaha?', to architecture khud is sawal ka jawab discoverable banati hai.

Category	Count
KSoR	1
MCP	20
ChatGPT Claude AI Agents / Custom Apps / Digital FTEs	3

KSoR search, embeddings, RAG, ya kisi bhi retrieval technique ko use kar sakta hai — lekin ye sab sirf **implementation details** hain. KSoR bunyadi tor par vector database, embedding service, RAG framework, chatbot, ya MCP wrapper nahi hai. In technologies se madad milti hai, lekin ye cheez khud KSoR nahi banati. Asal pehchan hai: **authoritative governed knowledge**.

KSoR ka sawal zyada bara hai: 'Kaunsi knowledge itni authoritative hai ke organization insaan aur AI agents ko is se operate karne ki ijazat deta hai?'

■ RAG, RSOIR ka hissa ho sakta hai. Lekin RSOIR sirf ek RAG system nahi hai — Governance, Authority, Provenance, Versioning, Review, Scope, Human Surface, Agent Surface — retrieval to inmein se ek chhota hissa hai.

Content Management System poochta hai: '*Content kaise banayen aur publish karein?*' KSoR poochta hai: '*Kaunsi knowledge authoritative, governed, traceable, aur insaan/agents ke bharose ke laiq hai?*' Content ek input hai — institutional knowledge asal asset hai, aur authority hi farq paida karti hai.

11. Governance Model

Source → Draft → Review → Approved → Authoritative KSoR

[illegible]

▼ ▼

Human Surface Agent Surface

Superseded / Retired

Regulated ya high-risk knowledge ke liye organizations extra controls laga sakti hain: named knowledge owners, approval requirements, effective dates, source citations, mandatory review periods, change records, conflict resolution, separation of duties, audit history. **KSoR architecture deta hai, governance policy khud organization ki zimmedari rehti hai.**

Knowledge Boundaries — teen misaalein

Type	Sawal	Kya hoga
In scope	'\$50,000 se zyada purchases par kaunsa approval threshold lagta hai?'	Agar policy mein jawab hai, KSoR jawab + source deta hai
Requires reasoning	'Kya is khaas purchase ko CFO approval chahiye?'	Governed rule ko operational facts (kisi doosre SoR se) ke saath jorna padta hai
Outside the KSoR	'Agle saal kaunsi approval policy aayegi?'	Agar approved nahi hai to system guess nahi karega, mana kar dega

■ Ye teenon misaalein trustworthy agentic systems ke liye zaroori hain — agent ko apni knowledge ki seemayen pata honi chahiye.

12. AI-Native Architecture Mein KSoR

Human / AI Worker

Agent

▼ ▼

KSoR Operational SoRs

(Policies/Methods) (CRM/ERP/HRIS, Ledger)

'kaise operate karein' 'abhi kya sach hai'

[illegible]

Decision → Action

KSoR agent ko batata hai **organization kaise operate karta hai**. Traditional SoR batata hai **abhi kya sach hai**. Dono mil kar reliable enterprise action ke live zaroori context dete hain.

13. Agent-First Development

KSoR aise development duniya ke liye design hua hai jahan coding agents zyada tar mechanical kaam khud karte hain. Scaffolded KSoR mein machine-readable instructions hoti hain jaise:

- 'Is policy ko KSoR mein add karo'
- 'In source documents ko governed Markdown mein convert karo'
- 'Har page par missing provenance check karo'
- 'Is section se ek quiz banao'
- 'KSoR validate karo' / 'Test suite chalo' / 'Release taiyar karo' / 'Deploy karo'

Isi liye repo do cheezein ban jata hai: **(1) khud knowledge ka artifact**, aur **(2) agents ke kaam karne ki working context** — taake subject-matter experts aur software engineers isi governed source par mil kar kaam kar saken.

Human-Readable by Default

KSoR ye zaroorat nahi karta ke saari knowledge sirf vector database mein gum ho jaye — canonical source hamesha inspectable rehta hai. Koi bhi isay khol sakta hai, padh sakta hai, diff kar sakta hai, review kar sakta hai, copy kar sakta hai — samajh sakta hai AI ko kya use karne ki ijazat hai.

Vendor-Free by Design

Aapka KSoR in sab tabdeeliyon se bach kar chalna chahiye: LLM providers, embedding models, vector stores, agent frameworks, cloud providers. Asal **durable asset governed knowledge hai** — baaki sab replaceable hona chahiye.

14. Deployment aur Firewall ke Peeche Kaam

ksor build se banaya hua human surface ek static site hota hai — Vercel, Netlify, GitHub Pages, static storage, ya khud ki internal servers par deploy ho sakta hai.

Kyunke KSoR mein aksar internal organizational knowledge hoti hai, isay is tarah design kiya gaya hai ke knowledge organization ke control mein rahe, website self-host ho sake, aur agent interface security boundary ke andar deploy ho sake. **KSoR kisi organization ko apni institutional knowledge third-party SaaS par publish karne par majboor nahi karta.**

15. Example Applications

Agent Factory KSoR

Ek shared methodology jo AI-native implementations mein use hoti hai: architecture, principles, FDE methodology, governance, agent patterns, evaluation standards, implementation methods, operating model.

Vertical KSoR

Ek profession/industry ke liye khaas knowledge — jaise Government Contract Accounting KSoR: accounting rules, FAR requirements, contract structures, indirect rates, revenue recognition, billing, compliance, workflows, decision criteria.



Agents dono ko mila kar domain-specific kaam kar sakte hain — dono ek hi ksoR SDK se banaye ja sakte hain.

KSoR Kya Replace Nahi Karta

KSoR aapke CRM, ERP, accounting system, HRIS, transactional database, data warehouse, ya operational APIs ki jagah nahi leta — ye systems apni operational state ke liye authoritative rehte hain. KSoR sirf woh **knowledge layer** add karta hai jo agents ko us state ko samajhne aur us par amal karne ke liye chahiye.

16. Design Goals (11 Usool)

- **Authoritative** — ek saaf canonical source ho
- **Governed** — knowledge ka malik aur controlled change ho
- **Traceable** — zaroori jawab evidence tak wapas jayen
- **Inspectable** — insaan dekh sakein agent kya padh raha hai
- **Portable** — kisi ek vendor mein phansi na ho
- **Agent-readable** — AI agents corpus ko programmatically use kar sakein
- **Human-readable** — log wahi knowledge browse aur samajh sakein
- **Versioned** — institutional truth mein history ho
- **Reproducible** — deployed KSoR ek khaas corpus/version se traceable ho
- **Composable** — kai KSoRs saath mil kar use ho sakein
- **Extensible** — organizations apni zaroorat ke hisaab se adapt kar sakein

17. Project Status

KSoR abhi active development mein hai — purane **VSOR** implementation se general **Knowledge System of Record** architecture ki taraf evolve ho raha hai. VSOR ne pehle hi kuch foundations prove ki thin: Markdown-based source, generated documentation sites, scaffolding, local dev, static builds, build

provenance, agent-oriented instructions, testing, deployment.

■ MCP-based agent surface abhi sirf **design** hui hai, implement nahi. License: **Apache 2.0**. npm package: @panaversity/ksor (command ksor hi rehta hai).

18. Ek Jumle Mein Poori Baat

■ **Traditional System of Record** AI agent ko batata hai business ke baare mein kya sach hai; **Knowledge System of Record** usay batata hai organization kya jaanta hai aur usay kaise operate karna chahiye. KSoR is knowledge ko authoritative, governed, traceable, human-readable, agent-readable, aur vendor-neutral banata hai.

Ye notes GitHub repo [panaversity/ksor](https://github.com/panaversity/ksor) ke README par mabni hain, apne alfaz mein tayyar kiye gaye. Asal aur live details ke liye dekhien: github.com/panaversity/ksor